

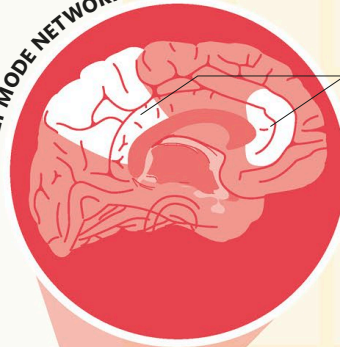
Creativity

We all get a creative spark from time to time, but what makes some of us more creative than others is linked to our connections and coordination between three different brain networks.

The science of creativity

Creativity—our ability to come up with new and useful ideas—is linked to three distinct brain networks: the default mode network, the salience network, and the central executive network. While these networks are linked, they are not typically active at the same time. However, fMRI studies of people asked to perform specific tasks show that people who can switch quickly between these networks at suitable moments have more creative responses to the task. The correlation is so strong, in fact, that a person's creativity can be predicted based on the strength of the connection between these networks.

DEFAULT MODE NETWORK

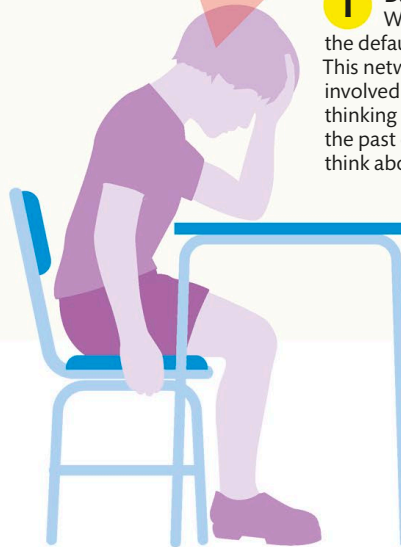


This network activates when the mind wanders

1

Daydreaming

When the mind wanders, the default mode network is active. This network includes brain regions involved with self-reflection, thinking of others, and considering the past or future—all things we think about when we daydream.



JAPANESE INVENTOR
SHUNPEI YAMAZAKI
HAS A REPORTED **5,255**
PATENTS TO HIS NAME



The creative brain

While genetics plays a role in creativity, other factors are also significant. Low levels of noradrenaline may support creativity as this neurotransmitter diverts inward-focused attention to external stimuli. While this might help our fight-or-flight response, creative ideas generally emerge from internal sources. Creativity may also require a strong knowledge base—composers, for example, tend to write their best work after decades of compositions.

